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Causal Representation Learning for Generalisable Recommendation

Yorgos Felekis[*] University of Warwick
Michael O'Riordan[*] Spotify
Oriol Corcoll Spotify
Ciarán M. Gilligan-Lee Spotify & University College London

Abstract

Predictive models trained on observational data often fail to generalise to the distributions they encounter when deployed, especially when the training data is a product of the system being optimised. Recommender systems are a canonical example: they are trained on interaction logs confounded by the deployed policy, past user behaviour, and platform filtering. As a result, the training distribution differs substantially from the candidate distribution scored at serving time, a gap that makes offline metrics unreliable predictors of online performance. We address the distribution shift problem with a method motivated by causal representation learning (CRL). We propose an information-theoretic disentanglement criterion and prove that its optimum excludes confounder-dependent non-causal information from the treatment representation. We then derive a tractable variational lower bound that makes the criterion optimisable from finite observational data alone. The scope of our method is narrower than that of much of the CRL literature, in that we target better generalisation under distribution shift, not full identification of all latent causal factors. This narrower target is what makes the method practical, requiring only the existing confounded logs, applying to any standard supervised model, and adding no inference-time cost. Our headline evaluation is an A/B test with millions of users on the music streaming platform Spotify, applied to a production ranker for personalised playlist generation. A CRL variant matched in capacity to the production baseline performed on par offline but delivered substantial online gains in listener engagement. Complementary evidence on a synthetic dataset with known causal structure and a semi-synthetic variant of the public KuaiRand recommendation dataset shows the same pattern: similar in-distribution performance to the baseline, gains under distribution shift. Across all three settings, adding our causal disentanglement objective yields better out-of-distribution generalisation.

1 Introduction

Predictive models deployed in feedback loops with the systems that produce their training data face a basic problem: the distribution they are trained on diverges from the one they are asked to score at serving time. In this work we focus on recommender systems, where the (user, item, context) tuples appearing in training logs are those a deployed policy already chose to surface, and are further filtered by user engagement. This phenomenon, broadly referred to as *exposure bias* in the recommendation literature Chen et al. [2023], means that at serving time the same model is asked to score a much broader and differently distributed candidate set. The gap between training and

[*]Equal contribution

Preprint.

serving-time distributions is widely understood to be a primary reason that offline ranking metrics are often unreliable predictors of online performance. Causal representation learning (CRL) Schölkopf et al. [2021] offers a principled solution. If a model's predictions rely only on causal mechanisms that generate the outcome, they should remain valid under shifts that leave those mechanisms intact, including the policy-induced shift between training and serving distributions. Classical CRL aims to recover latent causal factors with identifiability guarantees [Khemakhem et al., 2020]. Such identification requires additional assumptions or information [Locatello et al., 2019], which can be difficult to justify or obtain from logged observational recommendation data. A second strand applies CRL to recommendation directly, including in production. Wang et al. [2022] treat user feature drift as an intervention via a counterfactual VAE that infers latent user state and performs post-intervention inference at serving time. Corcoll et al. [2026] block-identify causal treatment representations via labelled contrastive pairs that stratify on confounders and outcomes. DCCL Zhao et al. [2023] and Rec4Ad Gao et al. [2023] disentangle specific bias mechanisms (popularity, sample-selection) with components tied to those mechanisms, and have been deployed online at Kuaishou and Taobao.

In this work, we assume that the relevant confounders are observed and present in the training data. In particular, we assume that confounders are encoded in the user, context, and item features that the deployed policy conditions on. In this case, full identification is more than what is needed to address the specific failure mode introduced by exposure bias. The spurious pathway can be suppressed directly by enforcing that the learned representation carries no information about the confounder beyond what the supervised target already requires (Figure 1). This narrower goal does not recover latent factors or enable counterfactual queries, but it does break the dependence that drives the offline-to-online gap, and it does so using only the existing observational training data. We instantiate this idea by learning a treatment representation that remains predictive of the outcome conditional on the observed confounder, while penalising mutual information between the learned representation and the confounder. The resulting criterion keeps the outcome-relevant causal content of the treatment and removes the confounder-dependent non-causal leakage that standard rankers may otherwise absorb. We make the following contributions:

- We propose an information-theoretic disentanglement criterion and prove that, at the population level, it favors representations that retain the causal content of the treatment and discard confounder-dependent non-causal information. We then derive a tractable variational lower bound that makes the criterion optimisable from finite observational data alone.
- We introduce a deployment-ready CRL recipe for robustness under distribution shift: a straightforward modification of any standard supervised model, trained on the existing confounded log, with no inference-time cost and no requirement for randomised or interventional data.
- We validate the method in three settings of increasing realism: a synthetic SCM, a semi-synthetic variant of the KuaiRand dataset Gao et al. [2022], and a live A/B test on a production personalised playlist generation surface with millions of users on the music streaming platform Spotify. Across all three settings, a capacity-matched CRL variant matches the baseline on in-distribution metrics and outperforms it when evaluation moves off the training distribution.

2 Method

2.1 Setting

We adopt the Structural Causal Model framework of Pearl [2009]. A causal model is a tuple$\mathcal{M} = (\mathbf{V}, \mathbf{U}, \mathcal{F}, P(\mathbf{U}))$, where $\mathbf{V}$ is the set of endogenous variables, $\mathbf{U}$ is the set of exogenous noise variables with joint distribution $P(\mathbf{U})$, and $\mathcal{F} = \{f_V\}_{V \in \mathbf{V}}$ is a collection of structural assignments $V := f_V(\mathrm{Pa}(V), U_V)$. In our setting, $\mathbf{V} = \{T, Y, Z\}$, where Z is an observed confounder (in recommendation, a user representation), T is the treatment to be ranked (an item), and Y is the outcome of interest. As illustrated in Figure 1, the observed treatment is an entangled mixing $T = m(T_C, T_{nC})$ of latent causal components T_C and non-causal components T_{nC}. We make two structural assumptions: **(A1)** $Y \perp T_{nC} \mid T_C, Z$ meaning that the outcome depends on the treatment only through its causal latents, given the confounder; **(A2)** $I(T_{nC}; Z \mid T_C) > 0$: which we call *unique spurious leakage*: the non-causal latents retain information about the confounder that is not already contained in the causal latents. This is the setting that arises in recommender systems whenever the deployed policy has correlated non-causal item-side factors with user-side signal, creating a spurious backdoor path $T \leftarrow T_{nC} \leftarrow Z \rightarrow Y$ that a standard predictor learning $\mathbb{E}[Y \mid T, Z]$ will absorb.

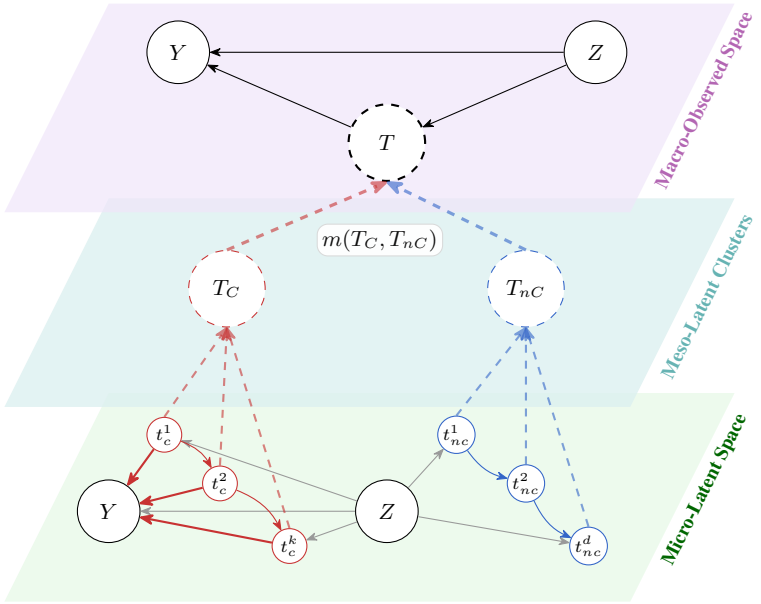

Figure 1: **Hierarchical Causal Entanglement.** *Bottom (Micro-Latent):* Only the red subset $\{t_c^i\}$ causally drives Y; the blue subset $\{t_{nc}^j\}$ contains spurious factors correlated with Z. *Middle (Meso-Latent):* Micro-variables abstract into causal (T_C) and non-causal (T_{nC}) clusters. *Top (Observed):* We observe only the entangled treatment $T = m(T_C, T_{nC})$, which creates the spurious backdoor path $T \leftarrow T_{nC} \leftarrow Z \rightarrow Y$ that standard models will exploit.

Such structures arise also in many causal representation learning settings where spurious correlations between treatment and confounders lead to biased models if not explicitly addressed.

While maximising $I(T; Y \mid Z)$ ensures the learned representation is predictive of the outcome, it may still preserve spurious signals as it does not penalise representations for carrying extra non-causal information. To address this, we introduce a more targeted objective that trades predictive utility against confounder dependence.

2.2 A New Objective for Causal Disentanglement

We propose to learn a representation $g(T)$ that maximizes the following:

$$J(g) = \underbrace{I(g(T); Y \mid Z)}_{\text{Utility}} - \lambda \underbrace{I(g(T); Z)}_{\text{Penalty}} \tag{1}$$

called the *Disentanglement Criterion*, where $\lambda \geq 0$ is a hyperparameter that controls the trade-off between predictive utility and confounder invariance. This objective balances two forces: the utility term rewards representations that are predictive of the outcome Y beyond Z, thereby encouraging the preservation of T_C; the penalty term discourages information shared with Z, thereby suppressing Z-correlated non-causal factors. Note that the penalty constrains the treatment representation $g(T)$, not the predictor's direct use of Z. The predictor still receives both $g(T)$ and Z. The objective discourages redundant confounder information in the treatment representation while rewarding information that predicts Y conditional on Z. Under our structural assumptions, the penalty term is exactly the channel through which T_{nC} leaks into the prediction via the spurious path $T_{nC} \leftarrow Z \rightarrow Y$. An equivalent bounded reparameterization is given in Appendix B. We first verify that the criterion ranks the ideal causal representation above the naive entangled one.

Proposition 1 (Ideal versus naive)**.** *Let $g^\star(T) = T_C$ and let $\tilde{g}(T) = T = (T_C, T_{nC})$ be a naive representation that retains both latent components. Under assumptions **A1** and **A2**, for any $\lambda > 0$,*

$$J(g^\star) > J(\tilde{g}). \tag{2}$$

Proof sketch. By the chain rule for CMI: $I(T_C, T_{nC}; Y \mid Z) = I(T_C; Y \mid Z) + I(T_{nC}; Y \mid T_C, Z)$. The second term is zero by **A1**, so the two representations have equal utility. For the penalty, $I(T_C, T_{nC}; Z) = I(T_C; Z) + I(T_{nC}; Z \mid T_C)$, and the second term is strictly positive by **A2**. Hence the naive representation incurs a strictly larger penalty while achieving the same utility. Therefore $J(g^\star) > J(\tilde{g})$. Full proof in Appendix A. □

Proposition 1 compares two fixed representations. We next formalise the more general mechanism through which the disentanglement criterion removes non-causal information. The observed treatment $T = m(T_C, T_{nC})$ may already lose information about T_C, as does a learned representation $g(T)$. We therefore define $\phi(T_C)$ as the information about T_C recoverable from $g(T)$ via a deterministic map r, i.e. $r(g(T)) = \phi(T_C)$. A *causally purified representation* is then $\bar{g} := r(g) = \phi(T_C)$. Because $\bar{g}$ is a deterministic function of g, the information in g decomposes exactly into the recovered causal content $\phi(T_C)$ and a residual beyond it. We assume that this residual carries no additional predictive information about Y given Z, but remains informative about the confounder, which are the representation-level counterparts of **A1** and **A2**: $I(g(T); Y \mid \phi(T_C), Z) = 0$, $I(g(T); Z \mid \phi(T_C)) > 0$. The following proposition shows that stripping the residual strictly improves J.

Proposition 2 (Causal purification)**.** *Let $g(T)$ be a representation satisfying the recoverability assumption, and define its causal purification as $\bar{g}(T) := r(g(T)) = \phi(T_C)$. Then, for any $\lambda > 0$,*

$$J(\bar{g}) > J(g). \tag{3}$$

Proof sketch. Since $\bar{g}$ is a deterministic function of g, we have $I(g; Z) = I(\bar{g}; Z) + I(g; Z \mid \bar{g})$.

By assumption, $I(g; Y \mid Z) = I(\phi(T_C); Y \mid Z) = I(\bar{g}; Y \mid Z)$, so the utility is preserved. The penalty gap is $I(g; Z) - I(\bar{g}; Z) = I(g; Z \mid \phi(T_C)) > 0$, which yields $J(\bar{g}) - J(g) = \lambda I(g; Z \mid \phi(T_C)) > 0$. Full proof in Appendix A. □

Remark 1. *Proposition 2 shows that the disentanglement criterion preserves the causal content available in the treatment representation while removing non-causal confounder-dependent information that is not needed for predicting Y. Hence, a maximising representation is purified with respect to the leakage targeted by the criterion and can be written as $g^\star(T) = \phi(T_C)$.*

However, the form of ϕ remains unspecified. Should it preserve all information in T_C (i.e. be invertible), or could a lossy compression be preferable? Since T_C is correlated with both Y (good for utility) and Z (costly for the penalty), any lossy ϕ reduces the penalty at the cost of some utility, with the trade-off governed by λ. To formalise this, let ϕ_{id} denote the identity (uncompressed) and let $\phi \in \mathcal{H}$ denote any non-invertible alternative transformation, where $\mathcal{H}$ is the class of all non-invertible lossy functions applicable to T_C. For each $\phi \in \mathcal{H}$ we define

$$\Delta U(\phi) := I(T_C; Y \mid Z) - I(\phi(T_C); Y \mid Z), \qquad \Delta P(\phi) := I(T_C; Z) - I(\phi(T_C); Z). \tag{4}$$

where $\Delta U(\phi)$ is the utility lost by compressing the causal component, while $\Delta P(\phi)$ is the reduction in confounding penalty gained by that compression. We assume a *no-free-lunch condition* according to which every nontrivial lossy transformation $\phi \in \mathcal{H}$ under consideration incurs a genuine utility loss and yields a genuine penalty reduction:

$$\Delta U(\phi) > 0 \text{ and } \Delta P(\phi) > 0. \tag{5}$$

We further define the *critical penalty weight* as the ratio of the two:

$$\Lambda(\phi) := \frac{\Delta U(\phi)}{\Delta P(\phi)}. \tag{6}$$

Proposition 3 (Lossless regime)**.** *Let $\lambda^\star := \inf_{\phi \in \mathcal{H}} \Lambda(\phi)$. Under the no-free-lunch condition* (5)*, for any $\lambda \in (0, \lambda^\star)$, the uncompressed representation T_C strictly dominates every lossy alternative $\phi(T_C)$ with $\phi \in \mathcal{H}$ under J; i.e. $J(T_C) > J(\phi(T_C))$, $\forall \phi \in \mathcal{H}$.*

Proof. Fix any $\phi \in \mathcal{H}$. Expanding the definition of J gives

$$I(T_C; Y \mid Z) - \lambda I(T_C; Z) > I(\phi(T_C); Y \mid Z) - \lambda I(\phi(T_C); Z) \iff \lambda < \frac{\Delta U(\phi)}{\Delta P(\phi)} = \Lambda(\phi)$$

Now suppose $\lambda \in (0, \lambda^\star)$. Since $\lambda < \lambda^\star = \inf_{\phi \in \mathcal{H}} \Lambda(\phi) \leq \Lambda(\phi)$ for every $\phi \in \mathcal{H}$, the condition $\lambda < \Lambda(\phi)$ holds uniformly, so $J(T_C) > J(\phi(T_C))$ for every $\phi \in \mathcal{H}$. Hence, throughout the regime $(0, \lambda^\star)$, the optimal representation preserves the full causal component. □

Proposition 3 shows that after non-causal leakage has been removed, sufficiently small penalty weights λ preserve the full causal component, and ϕ_{id} is optimal; compression of T_C becomes preferable only once the penalty reduction outweighs the corresponding utility loss. Finally, we provide a concrete sanity check for the preceding claims in Appendix C for a linear Gaussian SCM where all mutual-information terms can be computed in closed form: the appendix verifies that J strictly prefers the causal representation T_C over the naive entangled representation (T_C, T_{nC}), and illustrates the lossless regime by varying the amount of stochastic compression applied to T_C.

2.3 A Tractable Objective For Causal Disentanglement

The criterion in (1) cannot be optimised directly since both $I(g(T); Y \mid Z)$ and $I(g(T); Z)$ are unknown and intractable to compute from finite data. We therefore derive a tractable surrogate by replacing each term with a computable bound: a lower bound on the utility and an upper bound on the penalty so that the surrogate is a valid lower bound on $J(g)$.

Lower bound on the utility. For any conditional distribution $q(y \mid g(T), z)$, the standard variational bound on conditional mutual information (CMI) Barber and Agakov [2003] gives

$$I(g(T); Y \mid Z) \;\geq\; \mathbb{E}_{p(g(T),y,z)}\big[\log q(y \mid g(T), z)\big] + H(Y \mid Z), \tag{7}$$

which motivates learning q to approximate the true posterior $p(y \mid g(T), z)$. We instantiate this via a contrastive InfoNCE estimator van den Oord et al. [2018]: for each anchor $(g(T), y^+, z)$ we draw K negative outcomes $y_1^-, \ldots, y_K^-$ from $p(y \mid z)$ and define

$$\mathcal{L}_{\text{utility}}(\theta) := -\mathbb{E}\left[\log \frac{e^{f_\theta(g(T),\, y^+)/\tau}}{e^{f_\theta(g(T),\, y^+)/\tau} + \sum_{k=1}^{K} e^{f_\theta(g(T),\, y_k^-)/\tau}}\right]. \tag{8}$$

The InfoNCE loss directly lower-bounds the CMI van den Oord et al. [2018]:

$$I(g(T); Y \mid Z) \;\geq\; \log(K{+}1) - \mathcal{L}_{\text{utility}}(\theta) =: -\mathcal{L}_{\text{utility}}(\theta) + C_Y, \quad C_Y := \log(K{+}1). \tag{9}$$

Upper bound on the penalty. For any representation $g(T)$, and optimal InfoNCE critic $d_{\phi^\star}$, the NCE-CLUB estimator Liang et al. [2023] upper-bounds the mutual information between $g(T)$ and Z:

$$I(g(T); Z) \;\leq\; \underbrace{\mathbb{E}_{p(g(T),z)}\big[d_{\phi^\star}(g(T), z)\big] - \mathbb{E}_{p(g(T))p(z)}\big[d_{\phi^\star}(g(T), z)\big]}_{=:\, I_{\text{NCE-CLUB}}(g(T);\, Z;\, d_{\phi^\star})}. \tag{10}$$

The first term scores *positive* pairs $(g(T), z) \sim p(g, z)$, while the second scores *negative* pairs drawn independently from $p(g)p(z)$. The guarantee assumes an optimal critic recovering the log-density ratio $d_{\phi^\star}(g, z) = \log p(z \mid g) - \log p(z)$. In practice, we approximate this with a learnable critic $d_\phi(g(T), z)$, trained to discriminate joint samples $(g, z) \sim p(g, z)$ from independent samples $(g, z) \sim p(g)p(z)$. Further details are provided in Appendix D.

Substituting (9) and (10) into (1), yields a tractable variational lower bound for $J(g)$.

Theorem 1. *For any encoder g_θ and optimal InfoNCE critic $d_{\phi^\star}$,*

$$J(g_\theta) \;\geq\; -\mathcal{L}_{\text{utility}}(\theta) - \lambda\, I_{\text{NCE-CLUB}}(g_\theta(T);\, Z;\, d_{\phi^\star}) + C_Y. \tag{11}$$

Since $C_Y = \log(K{+}1)$ does not depend on $(\theta, \phi^\star)$, maximising this bound is equivalent to minimising

$$\mathcal{L}_{\text{final}}(\theta, \phi^\star) := \mathcal{L}_{\text{utility}}(\theta) + \lambda\, \mathcal{L}_{\text{penalty}}(\theta, \phi^\star), \quad \mathcal{L}_{\text{penalty}}(\theta, \phi^\star) := I_{\text{NCE-CLUB}}(g_\theta(T);\, Z;\, d_{\phi^\star}). \tag{12}$$

The resulting objective augments the standard ranking loss with an explicit mutual-information penalty that discourages dependence between the learned representation and the confounder, unlike IPM-based methods Shalit et al. [2017] which require explicit population matching.

2.4 Practical implementation

The tractable surrogate objective in Theorem 1 uses NCE-CLUB to upper-bound the mutual-information penalty. As described in Appendix D, direct optimisation alternates between fitting the critic with InfoNCE and updating the encoder against the resulting NCE-CLUB estimate, together

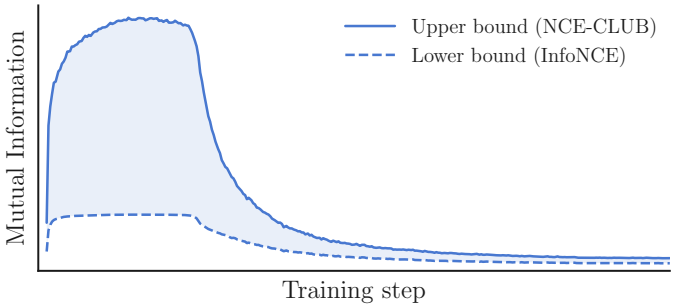

Figure 2: InfoNCE lower bound (dashed) and NCE-CLUB upper bound (solid) on $I(g(T); Z)$ during a production CRL training run. Gradient reversal acts on the InfoNCE bound. NCE-CLUB is tracked only as a diagnostic. Both rise during the critic warm-up, then decay as gradient reversal drives MI down. Axis values are omitted for confidentiality. As discussed in Section 2.4, a conservative reading of the bounds gives a reduction in MI of at least a factor of 5.

with the supervised utility loss. In our experiments, we instead find it convenient to apply gradient reversal [Ganin and Lempitsky, 2015] on the InfoNCE objective [van den Oord et al., 2018]. This allows the encoder and critic to be updated through a single backward pass, which was simpler to integrate into our existing production training pipeline. We emphasise that this choice is motivated by implementation convenience and is not a requirement of the proposed disentanglement criterion.

Of course, minimising an InfoNCE lower bound does not guarantee a decrease in the mutual information. We therefore track both the InfoNCE lower-bound estimate and the NCE-CLUB upper-bound estimate during training. Figure 2 shows a production run in which both estimates rise during critic warm-up and then decrease once gradient reversal becomes active. In particular, the NCE-CLUB estimate decreases even though it is not directly optimised. To quantify the MI reduction relative to the non-CRL baseline model, we compare the final CRL upper bound, U_{CRL}, with the baseline lower bound, L_{base}. Writing I_{CRL} and I_{base} for their respective actual MI, the validity conditions of the bounds give $I_{\mathrm{CRL}} \leq U_{\mathrm{CRL}}$ and $L_{\mathrm{base}} \leq I_{\mathrm{base}}$. In our production setup, this diagnostic gives $U_{\mathrm{CRL}} \leq L_{\mathrm{base}}/5$, which implies $I_{\mathrm{CRL}} \leq U_{\mathrm{CRL}} \leq \frac{1}{5} L_{\mathrm{base}} \leq \frac{1}{5} I_{\mathrm{base}}$. Therefore, assuming valid bounds, we can conclude that the MI has decreased by at least a factor of 5 relative to the non-CRL model. If this NCE-CLUB diagnostic does not support a reduction in MI, we recommend direct optimisation of the NCE-CLUB penalty following Appendix D.

An excessively strong penalty can remove information useful for prediction. Proposition 3 characterises the ideal lossless regime, but its population threshold is not directly observable. Moreover, the scale of λ depends on the parameterisation of the utility and penalty losses, the critic, and the application. We therefore treat λ as a standard training hyperparameter and suggest using in-distribution validation performance as a safeguard against over-regularisation. We recommend increasing λ from zero and selecting the strongest regularisation that keeps the validation metric within an application-specific tolerance of the $\lambda = 0$ baseline. The MI diagnostics provide additional guidance on whether the penalty is adequately reducing confounder dependence. Consistent with this approach, our CRL and non-CRL models remain close on in-distribution metrics, while the CRL models improve under distribution shift.

3 Related work

Treatment-effect estimation with neural networks. CFR Shalit et al. [2017] augments TARNet with an integral-probability-metric penalty that balances covariate representations across treatment groups. CEVAE Louizos et al. [2017] addresses unobserved confounders with a deep generative model using proxies. See also Jeunen et al. [2022], Van Goffrier et al. [2023], Fawkes et al. [2025] for alternate methods of dealing with unobserved confounders. Dragonnet Shi et al. [2019] adds a propensity-prediction head to encourage representations sufficient for treatment-effect identification. Deep Twin Networks Vlontzos et al. [2023, 2026], Reynaud et al. [2022] adapt neural networks to answer counterfactual questions Richens et al. [2020], Pearl [2009] by ensuring their architecture matches the twin network Pearl [2009], Graham et al. [2019] of the causal structure in question.

Related work studies structured treatments Kaddour et al. [2021], Harada and Kashima [2021] and effects of linguistic properties Pryzant et al. [2021]. GraphITE penalises dependence between treatment and individual representations using HSIC. We instead use an information-theoretic criterion with a causal-purification analysis. Corcoll et al. [2026] learn causal treatment representations using contrastive pairs stratified on confounders and outcomes, whereas our approach does not require constructing stratified contrastive pairs.

Debiasing causal recommendation. Wang et al. [2022] address shifts in user attributes with a causal variational autoencoder that predicts preferences under updated attributes while reducing the influence of outdated interaction history. Debiasing approaches include IPS reweighting Schnabel et al. [2016], Joachims et al. [2017], pseudo-rating-based asymmetric tri-training Saito [2020], instrumental-variable methods such as IViDR Deng et al. [2024], proxy-based latent-confounder identification in iDCF Zhang et al. [2023], and causal adjustment Wang et al. [2020], Zhang et al. [2021]. CausE Bonner and Vasile [2018] couples embeddings learned from biased and unbiased exposure data, while DICE Zheng et al. [2021] and DCCL Zhao et al. [2023] disentangle interest and conformity. Rec4Ad Gao et al. [2023] augments advertising logs with organic recommendation samples and learns aligned, decorrelated representations. Our method instead regularises treatment-confounder dependence within the existing training log.

Adversarial MI estimation and invariance. The practical estimator combines gradient reversal Ganin and Lempitsky [2015] with the InfoNCE lower bound van den Oord et al. [2018]. Alternative MI estimators include MINE Belghazi et al. [2018] and the CLUB upper bound Cheng et al. [2020]; NCE-CLUB Liang et al. [2023] provides the upper-bound machinery used in our diagnostics (Appendix D). MI-based representation objectives are also at the core of Deep InfoMax Hjelm et al. [2019]. The disentanglement criterion $J(g)$ is complementary to invariance-based approaches, including invariant causal prediction Peters et al. [2016] and invariant risk minimisation Arjovsky et al. [2019]. These exploit multiple environments, whereas we penalise dependence on an observed confounder without requiring an environment partition.

Causal bottlenecks and abstractions. Our objective is related to causal information-bottleneck methods, which retain task-relevant causal information while discarding irrelevant variation. The Causal Information Bottleneck Simoes et al. [2025] learns causal variable abstractions by compressing an input while preserving causal control over a target, and Structural Causal Bottleneck Models Bing et al. [2026] assume that causal effects between high-dimensional SCM variables are mediated by lower-dimensional bottlenecks of their causes. These perspectives connect to causal abstraction Rubenstein et al. [2017], which seeks coarser variables or models that are interventionally consistent with the original system, together with methods that learn such maps Zennaro et al. [2023], Felekis et al. [2024], Kekić et al. [2024], Xia and Bareinboim [2024], Felekis et al. [2026]. In the language of Figure 1, such methods would ask whether the entangled $T = m(T_C, T_{nC})$ admits a coarser description, ideally T_C, that preserves interventional consistency. Our setting is adjacent but deliberately weaker: with m fixed and unobserved, we learn a map $T \mapsto g(T)$ sufficient for the recoverable causal content $\phi(T_C)$, while removing the Z-correlated non-causal leakage carried by T_{nC}. We do not identify T_C, learn m, or verify model-level consistency; we learn a representation that retains the causal information needed for prediction.

4 Experiments

We evaluate the proposed method in three settings of increasing realism. The synthetic SCM (§4.1) shows that the method successfully isolates the causal components of the treatment representation in a simple setup where the full data generation process is known. The semi-synthetic KuaiRand experiment (§4.2) uses real user and video features, with a synthetic nonlinear outcome function to test robustness to a controlled change in non-causal feature assignment. The online A/B test (§4.3) verifies that this robustness translates into measurable online gains on a deployed production model at Spotify.

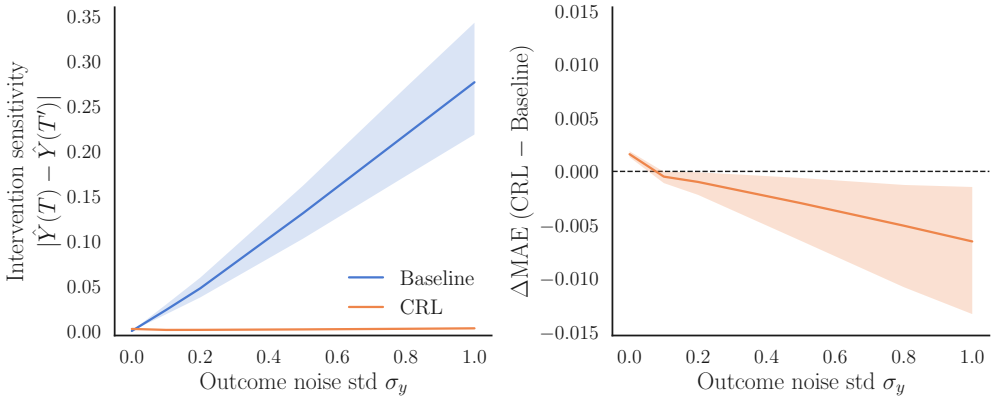

Figure 3: Synthetic SCM, sweep over outcome-noise scale $\sigma_Y \in [0, 1]$ across 2,000 seeds per noise level. Lines are per-seed medians; shaded bands are the inter-quartile range across seeds. **Left:** intervention sensitivity $|\hat{Y}(T) - \hat{Y}(T')|$, where T' resamples the non-causal coordinates of T; lower is better, with 0 indicating that the model has not absorbed any signal from T_{nC}. **Right:** paired $\Delta\text{MAE} = \text{MAE}_{\text{CRL}} - \text{MAE}_{\text{baseline}}$ per seed, centered on the zero line. The CRL penalty does not degrade predictive performance, and is in fact slightly better at higher noise levels.

4.1 Synthetic SCM

We instantiate a linear additive SCM with a confounder $Z \in \mathbb{R}^{10}$, a treatment $T \in \mathbb{R}^{10}$ decomposable into $d_c = 5$ causal and 5 non-causal dimensions, and an outcome Y that depends on T_C and Z but not T_{nC}. Full data-generating equations are given in Appendix E. Since Y is invariant to interventions on T_{nC}, a model that has correctly isolated the causal part of the treatment should also be invariant to those interventions. We measure the *intervention sensitivity* $|\hat{Y}(T) - \hat{Y}(T')|$ where T' has its non-causal coordinates resampled, and the *prediction MAE* on Y. In Figure 3 we sweep the outcome noise σ_Y from 0 to 1, and show how the baseline predictor is sensitive to interventions on the T_{nC} non-causal signal, whereas the CRL variant's intervention sensitivity sits essentially at zero across the entire range. The MAE is matched between the two methods.

4.2 Semi-Synthetic KuaiRand

In this section, we evaluate our method in a semi-synthetic setup using real user and video features from KuaiRand-Pure [Gao et al., 2022]. We generate synthetic clicks with a fixed nonlinear function of user features Z, video features T_C, and their interactions. We introduce a non-causal feature T_{nC} and append it to the video input. In the source distribution, this feature is correlated with Z, providing a shortcut which the model can exploit for prediction. In the target distribution, T_{nC} is assigned uniformly at random. The source and target evaluations use the same held-out users, videos, and synthetic clicks, with only the shortcut assignment changing.

We use the DeepFM [Guo et al., 2017] architecture with user and video features and learned ID embeddings. DeepFM is a standard baseline in the recommendation literature [see eg. Wang et al., 2021], and has been applied to the KuaiRand dataset [Wang et al., 2024, Huang et al., 2024]. Our CRL variant adds an InfoNCE critic between the treatment and user representations, with gradient reversal applied to the treatment representation. The predictor architecture and capacity are the same as the baseline. Details of the data-generation, architecture, training setup, and metrics are in Appendix F.

We compare our CRL approach with the baseline, DCCL [Zhao et al., 2023], and two oracle methods. DCCL is similar to our approach in that it adds contrastive losses during training without requiring target-domain data. However, it targets popularity and conformity through user-conditioned positive and negative interaction pairs. Our objective does not specify a particular bias mechanism and does not use user or outcome-specific positive/negative pair construction. We further note that DCCL's specific bias mechanism and pair construction are too restrictive for our production application. Oracle IPS uses the exact target-to-source T_{nC} probability ratio. Oracle T_{nC} randomization resamples each

Table 1: Semi-synthetic KuaiRand results over 100 seeds. AUC entries report means ± 95% confidence intervals. Ranking metrics report target means. Source and target use the same held-out rows, changing only T_{nC} assignment. Oracle methods receive information about the true assignment mechanism. Our CRL model improves all target metrics over the baseline and approaches oracle T_{nC} randomization performance.

Method	Source AUC	Target AUC	Recall@5	NDCG@5
Baseline	0.7249 ± 0.0004	0.6914 ± 0.0011	0.5697	0.4501
DCCL	0.7248 ± 0.0004	0.6912 ± 0.0011	0.5699	0.4501
Oracle IPS	0.7079 ± 0.0008	0.7078 ± 0.0008	0.5850	0.4678
Oracle T_{nC} randomization	0.7243 ± 0.0004	0.7243 ± 0.0004	0.5884	0.4717
CRL (ours)	0.7218 ± 0.0005	0.7212 ± 0.0004	0.5853	0.4676

Table 2: Online A/B test results on the personalised playlist generation surface. Percentage change of the CRL variant relative to the production baseline, with 95% confidence intervals.

Metric	% change	95% CI
Track Streams	$+0.75\%$	$[+0.54\%, +0.96\%]$
Track Skips	-0.61%	$[-0.74\%, -0.47\%]$
Minutes Played	$+0.50\%$	$[+0.27\%, +0.73\%]$

training example's T_{nC} uniformly. Both oracle approaches rely on knowledge of the synthetic assignment mechanism.

In Table 1, we show that the baseline is sensitive to distribution shifts in the non-causal shortcut T_{nC}, with AUC degrading between the source and target distributions. DCCL performs similarly to the baseline in this setup. Oracle IPS shows little or no degradation between source and target, but large weights and low effective sample size reduce overall performance in this setup. The baseline and CRL models are close on the source distribution, but CRL is robust to the distribution shift, substantially reducing the AUC degradation between source and target. CRL outperforms the baseline across all target metrics, and approaches the performance of the oracle shortcut randomization. This pattern of similar performance on the source distribution combined with gains on the target distribution is similar to what Taori et al. [2020] call *effective robustness*: improvement under shift not predicted by in-distribution performance alone. In the online A/B test (Section 4.3) we show how this improved generalisation ability translates to real gains in listener engagement.

4.3 Online A/B test on personalised playlist generation

We evaluate real-world impact through an online A/B test with millions of users on a production personalised playlist generation surface at Spotify. The test compares two versions of the production session-ranking model: the deployed baseline and a capacity-matched CRL variant trained with the disentanglement objective of §2. The CRL variant uses an adversarial loss to decorrelate the model's internal representation of the candidate track from its representations of session- and user-level context, mirroring the recipe used in the KuaiRand experiment. Both variants perform similarly on in-distribution offline metrics, but the CRL variant performs better online. Table 2 reports the headline engagement metrics, measured over two weeks during the test. The CRL variant increases track streams by $+0.75\%$ and minutes played by $+0.50\%$ while decreasing track skips by -0.61%, with all three changes significant at the 95% level. For context, these gains are of the same order of magnitude as the improvements that the current production ranker, itself a ground-up rebuild, delivered over its mature predecessor.

4.4 Discussion

Across all three settings, in-distribution performance remains unchanged while performance under distribution shift improves. The synthetic and semi-synthetic experiments demonstrate robustness to known non-causal treatment variation, and the production A/B test demonstrates online gains in listener engagement. Our theoretical analysis assumes that the relevant confounding information

is observed in Z. The penalty cannot remove dependence transmitted only through unobserved confounders. In production, the available user and session context almost surely does not include all relevant confounding factors. The observed gains nevertheless suggest that an informative but incomplete Z can still improve generalisation. Combining the criterion with proximal causal inference is a possible extension, requiring suitable proxies and additional identifying assumptions.

5 Conclusion

We proposed a CRL-motivated method for making predictive models more robust under the distribution shift characteristic of on-policy recommendation. We introduced an information-theoretic disentanglement criterion, characterised its properties, and derived a tractable training objective. The method uses existing observational logs and adds no inference-time cost. Synthetic and semi-synthetic experiments demonstrate improved robustness under controlled shifts, while a large-scale production A/B test showed significant gains in listener engagement.

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A Proofs for §2

Throughout, we use the structural assumptions introduced in §2. The observed treatment decomposes as

$$T = m(T_C, T_{nC}),$$

the outcome is conditionally independent of the non-causal latents,

$$\textbf{(A1)} \qquad Y \perp T_{nC} \mid (T_C, Z),$$

and the non-causal latents contain residual confounder information beyond the causal latents,

$$\textbf{(A2)} \qquad I(T_{nC}; Z \mid T_C) > 0.$$

Proof of Proposition 1

We compare the ideal representation $g^\star(T) = T_C$ with the naive representation $\tilde{g}(T) = T = (T_C, T_{nC})$.

Utility. By the chain rule for conditional mutual information,

$$\begin{aligned} I(\tilde{g}(T); Y \mid Z) &= I(T_C, T_{nC}; Y \mid Z) \\ &= I(T_C; Y \mid Z) + I(T_{nC}; Y \mid T_C, Z). \end{aligned}$$

The second term is zero by **A1**. Hence

$$I(\tilde{g}(T); Y \mid Z) = I(T_C; Y \mid Z) = I(g^\star(T); Y \mid Z).$$

Penalty. Again, by the mutual-information chain rule,

$$\begin{aligned} I(\tilde{g}(T); Z) &= I(T_C, T_{nC}; Z) \\ &= I(T_C; Z) + I(T_{nC}; Z \mid T_C). \end{aligned}$$

By **A2**, $I(T_{nC}; Z \mid T_C) > 0$. Therefore

$$I(\tilde{g}(T); Z) > I(T_C; Z) = I(g^\star(T); Z).$$

Equivalently, using the entropy chain rule,

$$\begin{aligned} I(T_C, T_{nC}; Z) &= H(T_C, T_{nC}) - H(T_C, T_{nC} \mid Z) \\ &= \big[H(T_C) + H(T_{nC} \mid T_C)\big] - \big[H(T_C \mid Z) + H(T_{nC} \mid T_C, Z)\big] \\ &= I(T_C; Z) + I(T_{nC}; Z \mid T_C). \end{aligned}$$

Thus, the excess penalty of the naive representation is exactly the unique spurious leakage term.

Conclusion. The two representations have equal utility, but the naive representation has a strictly larger penalty. Hence

$$\begin{aligned} J(g^\star) - J(\tilde{g}) &= \big[I(g^\star; Y \mid Z) - I(\tilde{g}; Y \mid Z)\big] + \lambda\big[I(\tilde{g}; Z) - I(g^\star; Z)\big] \\ &= \lambda\, I(T_{nC}; Z \mid T_C) > 0. \end{aligned}$$

Therefore $J(g^\star) > J(\tilde{g})$. □

Proof of Proposition 2

Consider a representation satisfying the recoverability assumption:

$$g(T) = h\big(\phi(T_C), \ell(T_{nC})\big),$$

where the causal component $\phi(T_C)$ is deterministically recoverable from $g(T)$. That is, there exists a map r such that

$$r(g(T)) = \phi(T_C).$$

Define the purified representation by

$$\bar{g}(T) := r(g(T)) = \phi(T_C).$$

Thus $\bar{g}$ is a deterministic function of g. At the representation level, we assume that the residual information in $g(T)$ beyond the recovered causal component contains no additional outcome information but does contain residual confounder information:

$$\textbf{(R1)} \qquad I(g(T); Y \mid \phi(T_C), Z) = 0,$$

and

$$\textbf{(R2)} \qquad I(g(T); Z \mid \phi(T_C)) > 0.$$

Utility preservation. Since $\bar{g} = \phi(T_C)$ is a deterministic function of g, the pair $(g, \bar{g})$ contains the same information as g. Hence

$$I(g; Y \mid Z) = I(g, \bar{g}; Y \mid Z).$$

By the chain rule for conditional mutual information,

$$I(g, \bar{g}; Y \mid Z) = I(\bar{g}; Y \mid Z) + I(g; Y \mid \bar{g}, Z).$$

Since $\bar{g} = \phi(T_C)$, the second term is

$$I(g; Y \mid \bar{g}, Z) = I(g; Y \mid \phi(T_C), Z),$$

which is zero by **R1**. Therefore

$$I(g; Y \mid Z) = I(\bar{g}; Y \mid Z).$$

Thus purification preserves the utility term.

Penalty reduction. Again, since $\bar{g}$ is a deterministic function of g,

$$I(g; Z) = I(g, \bar{g}; Z).$$

Applying the mutual-information chain rule,

$$I(g, \bar{g}; Z) = I(\bar{g}; Z) + I(g; Z \mid \bar{g}).$$

Using $\bar{g} = \phi(T_C)$, we obtain

$$I(g; Z) - I(\bar{g}; Z) = I(g; Z \mid \bar{g}) = I(g; Z \mid \phi(T_C)).$$

By **R2**, this quantity is strictly positive. Hence

$$I(\bar{g}; Z) < I(g; Z).$$

Objective comparison. Combining utility preservation and penalty reduction,

$$\begin{aligned} J(\bar{g}) - J(g) &= \big[I(\bar{g}; Y \mid Z) - I(g; Y \mid Z)\big] + \lambda\big[I(g; Z) - I(\bar{g}; Z)\big] \\ &= \lambda\, I(g; Z \mid \phi(T_C)) > 0. \end{aligned}$$

Therefore $J(\bar{g}) > J(g)$ for every $\lambda > 0$. □

General Predictive Purification Result

The purification result in the main text assumes that the causal component $\phi(T_C)$ is recoverable from the representation. We now show that a more general purification principle holds without this assumption.

Let

$$G := g(T)$$

be an arbitrary representation. We call $\bar{G} = s(G)$ a *predictive purification* of G if it is a deterministic function of G satisfying

$$Y \perp G \mid (\bar{G}, Z).$$

In words, once $\bar{G}$ and Z are known, the remaining information in G does not provide additional predictive information about Y.

Existence. Under standard regularity conditions, such a statistic exists. One canonical construction is the conditional response profile

$$s(G) := \big\{p(Y \mid G, Z = z) : z \in \mathcal{Z}\big\}.$$

This object records, for each value of G, the conditional law of Y across values of the confounder. By construction,

$$p(Y \mid G, Z) = p(Y \mid s(G), Z),$$

which implies

$$Y \perp G \mid (s(G), Z).$$

This statistic may be infinite-dimensional and is used only as a population-level object.

Proposition 4 (General predictive purification)**.** *Let $G = g(T)$ be any representation, and let $\bar{G} = s(G)$ be a predictive purification. Then, for any $\lambda > 0$,*

$$J(\bar{G}) \geq J(G),$$

with exact improvement

$$J(\bar{G}) - J(G) = \lambda I(G; Z \mid \bar{G}).$$

The inequality is strict whenever $I(G; Z \mid \bar{G}) > 0$.

Proof. Since $\bar{G} = s(G)$ is a deterministic function of G,

$$I(G; Y \mid Z) = I(G, \bar{G}; Y \mid Z).$$

By the chain rule,

$$I(G, \bar{G}; Y \mid Z) = I(\bar{G}; Y \mid Z) + I(G; Y \mid \bar{G}, Z).$$

The second term is zero by predictive sufficiency, so

$$I(G; Y \mid Z) = I(\bar{G}; Y \mid Z).$$

For the penalty,

$$I(G; Z) = I(G, \bar{G}; Z) = I(\bar{G}; Z) + I(G; Z \mid \bar{G}).$$

Thus

$$I(G; Z) - I(\bar{G}; Z) = I(G; Z \mid \bar{G}) \geq 0.$$

Combining,

$$J(\bar{G}) - J(G) = \lambda I(G; Z \mid \bar{G}) \geq 0.$$

□

Connection to the causal setting. The result above is purely information-theoretic and applies to any representation. Its causal interpretation follows from assumption **A1**. For the full entangled latent representation

$$G = (T_C, T_{nC}),$$

we have

$$p(Y \mid T_C, T_{nC}, Z) = p(Y \mid T_C, Z),$$

so the predictive purification depends only on the causal component T_C, up to outcome-equivalence. Thus, in the SCM considered in this work, the population target of purification can be written as

$$g(T) = \phi(T_C).$$

Relation to the main result. Proposition 4 shows that the disentanglement criterion removes information that is irrelevant for predicting $Y \mid Z$ but still informative about the confounder, without requiring any structural assumptions on g. The main-text result strengthens this conclusion by imposing a recoverability assumption, which allows the purified representation to be identified explicitly as a function of the causal component T_C.

B Reparameterisation of the Disentanglement Criterion

The objective in (1),

$$J(g) = I(g(T); Y \mid Z) - \lambda I(g(T); Z),$$

uses the unbounded penalty weight $\lambda \in [0, \infty)$. It is often more convenient to work with an equivalent bounded parameter $\gamma \in [0, 1]$.

Reparameterisation. For $\lambda > 0$, maximising $J(g)$ is equivalent to minimising

$$-J(g) = \lambda I(g(T); Z) - I(g(T); Y \mid Z).$$

Dividing by the positive constant $1 + \lambda$ does not change the minimiser, so the same solutions are obtained by minimising

$$\frac{\lambda}{1+\lambda} I(g(T); Z) - \frac{1}{1+\lambda} I(g(T); Y \mid Z).$$

Define

$$\gamma := \frac{1}{1+\lambda} \in (0, 1),$$

so that $1 - \gamma = \lambda/(1 + \lambda)$. The equivalent bounded objective is

$$L_\gamma(g) := (1-\gamma) I(g(T); Z) - \gamma I(g(T); Y \mid Z), \tag{13}$$

which we minimise over g. The limiting case $\lambda = 0$ corresponds to $\gamma = 1$, i.e. pure utility maximisation. Conversely, $\lambda \to \infty$ corresponds to $\gamma \to 0$, i.e. pure penalty minimisation.

Optimality condition under γ. Using the same notation as Proposition 3, compare the uncompressed representation T_C with a lossy alternative $\phi(T_C)$. The uncompressed representation is preferred under L_γ when

$$L_\gamma(T_C) < L_\gamma(\phi(T_C)).$$

Expanding this inequality gives

$$(1-\gamma)\Delta P(\phi) - \gamma \Delta U(\phi) < 0,$$

or equivalently

$$\gamma > \frac{\Delta P(\phi)}{\Delta U(\phi) + \Delta P(\phi)} =: \Gamma(\phi).$$

Thus $\Gamma(\phi)$ is the critical utility weight above which the full causal component is preferred to the lossy compression ϕ.

Proposition 5 (High-utility regime under γ)**.** *Let $\mathcal{H}$, $\Delta U(\phi)$, and $\Delta P(\phi)$ be as in Proposition 3, and assume the no-free-lunch condition $\Delta U(\phi) > 0$ and $\Delta P(\phi) > 0$ for every $\phi \in \mathcal{H}$. Define*

$$\gamma^\star := \sup_{\phi \in \mathcal{H}} \Gamma(\phi) = \sup_{\phi \in \mathcal{H}} \frac{\Delta P(\phi)}{\Delta U(\phi) + \Delta P(\phi)}.$$

Then for any $\gamma \in (\gamma^\star, 1)$, the uncompressed representation T_C strictly dominates every lossy alternative $\phi(T_C)$, $\phi \in \mathcal{H}$, under the objective L_γ.

Proof. Fix any $\phi \in \mathcal{H}$. The condition $L_\gamma(T_C) < L_\gamma(\phi(T_C))$ holds if and only if $\gamma > \Gamma(\phi)$. By definition of the supremum,

$$\Gamma(\phi) \le \gamma^\star \qquad \forall\, \phi \in \mathcal{H}.$$

Hence, for any $\gamma > \gamma^\star$,

$$\gamma > \Gamma(\phi) \qquad \forall\, \phi \in \mathcal{H}.$$

Therefore $L_\gamma(T_C) < L_\gamma(\phi(T_C))$ uniformly over $\mathcal{H}$, so the uncompressed causal representation strictly dominates all lossy alternatives. □ □

Correspondence with Proposition 3. The λ- and γ-parameterisations describe the same trade-off. Since

$$\gamma = \frac{1}{1+\lambda},$$

the condition

$$\lambda < \Lambda(\phi) = \frac{\Delta U(\phi)}{\Delta P(\phi)}$$

is equivalent to

$$\gamma > \frac{\Delta P(\phi)}{\Delta U(\phi) + \Delta P(\phi)} = \Gamma(\phi).$$

Thus the low-λ lossless regime in Proposition 3 is the same as the high-γ regime in Proposition 5.

C Toy Example

We give a closed-form sanity check for the Disentanglement Criterion:

$$J(g) = I(g(T); Y \mid Z) - \lambda I(g(T); Z)$$

in a linear additive Gaussian model. The treatment consists of causal and non-causal components, but only the causal ones affects the outcome. Consider the following SCM:

$$Z = \epsilon_Z, \tag{14}$$
$$T_C = \alpha Z + \epsilon_C, \tag{15}$$
$$T_{nC} = \beta Z + \epsilon_N, \tag{16}$$
$$Y = \rho T_C + \delta Z + \epsilon_Y, \tag{17}$$

where all noise variables are mutually independent and Gaussian:

$$\epsilon_Z \sim \mathcal{N}(0, \sigma_Z^2), \quad \epsilon_C \sim \mathcal{N}(0, \sigma_C^2), \quad \epsilon_N \sim \mathcal{N}(0, \sigma_N^2), \quad \epsilon_Y \sim \mathcal{N}(0, \sigma_Y^2).$$

The observed treatment is

$$T = (T_C, T_{nC}).$$

By construction, the outcome depends on the treatment only through the causal component:

$$Y \perp T_{nC} \mid (T_C, Z).$$

Moreover, under non-degenerate parameters, e.g. $\beta \neq 0$, $\sigma_Z^2 > 0$, $\sigma_C^2 > 0$, and $\sigma_N^2 > 0$, the non-causal component contains residual information about the confounder beyond the causal component:

$$I(T_{nC}; Z \mid T_C) > 0.$$

Indeed, since the variables are jointly Gaussian,

$$I(T_{nC}; Z \mid T_C) = \frac{1}{2} \log \frac{\mathrm{Var}(T_{nC} \mid T_C)}{\mathrm{Var}(T_{nC} \mid Z, T_C)}.$$

Now

$$\mathrm{Var}(T_{nC} \mid Z, T_C) = \sigma_N^2,$$

whereas

$$\mathrm{Var}(T_{nC} \mid T_C) = \beta^2 \mathrm{Var}(Z \mid T_C) + \sigma_N^2,$$

with

$$\mathrm{Var}(Z \mid T_C) = \frac{\sigma_Z^2 \sigma_C^2}{\alpha^2 \sigma_Z^2 + \sigma_C^2}.$$

Therefore,

$$I(T_{nC}; Z \mid T_C) = \frac{1}{2} \log \left(1 + \frac{\beta^2 \sigma_Z^2 \sigma_C^2}{\sigma_N^2 (\alpha^2 \sigma_Z^2 + \sigma_C^2)} \right) > 0.$$

Thus the Gaussian SCM satisfies assumptions **A1** and **A2** from the main text. Consequently, we now compare the ideal causal representation with the naive entangled representation:

$$g_1(T) = T_C, \quad g_2(T) = T = (T_C, T_{nC}).$$

For scalar jointly Gaussian variables,

$$I(U; V \mid W) = \frac{1}{2} \log \frac{\mathrm{Var}(V \mid W)}{\mathrm{Var}(V \mid U, W)}.$$

Conditioned on Z,

$$Y = (\rho\alpha + \delta) Z + \rho \epsilon_C + \epsilon_Y,$$

so

$$\mathrm{Var}(Y \mid Z) = \rho^2 \sigma_C^2 + \sigma_Y^2.$$

Conditioned on (T_C, Z),

$$Y = \rho T_C + \delta Z + \epsilon_Y,$$

and therefore

$$\mathrm{Var}(Y \mid T_C, Z) = \sigma_Y^2.$$

Hence

$$I(T_C; Y \mid Z) = \frac{1}{2} \log \left(\frac{\rho^2 \sigma_C^2 + \sigma_Y^2}{\sigma_Y^2} \right) \tag{18}$$

$$= \frac{1}{2} \log \left(1 + \frac{\rho^2 \sigma_C^2}{\sigma_Y^2} \right). \tag{19}$$

Since $Y \perp T_{nC} \mid (T_C, Z) \implies I(T_C, T_{nC}; Y \mid Z) = I(T_C; Y \mid Z)$. So the ideal and naive representations have identical utility.

For the penalty term,

$$I(T_C; Z) = \frac{1}{2} \log \left(\frac{\alpha^2 \sigma_Z^2 + \sigma_C^2}{\sigma_C^2} \right) = \frac{1}{2} \log \left(1 + \frac{\alpha^2 \sigma_Z^2}{\sigma_C^2} \right).$$

By the mutual-information chain rule,

$$I(T_C, T_{nC}; Z) = I(T_C; Z) + I(T_{nC}; Z \mid T_C).$$

The second term is strictly positive by the calculation above, so

$$I(g_2(T); Z) > I(g_1(T); Z).$$

Therefore, for any $\lambda > 0$,

$$J(g_1) - J(g_2) = \lambda \big[I(g_2(T); Z) - I(g_1(T); Z) \big] \tag{20}$$

$$= \lambda I(T_{nC}; Z \mid T_C) > 0. \tag{21}$$

Thus, in this Gaussian SCM, the criterion strictly prefers the causal representation T_C over the naive entangled representation (T_C, T_{nC}). Figure 4 shows the same effect empirically in a multivariate simulation where the representation progressively removes coordinates of T_{nC}. Retaining more non-causal information leaves the utility unchanged but increases the confounding penalty, so the corresponding objective values decrease more rapidly as λ grows.

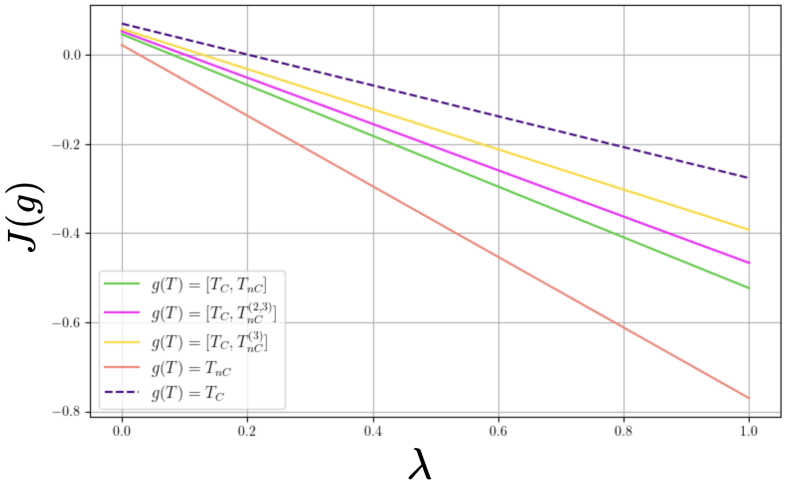

Figure 4: **Progressive removal of non-causal leakage.** The objective $J(g)$ is plotted as a function of the penalty weight λ for representations retaining different amounts of the non-causal component T_{nC}. Since T_{nC} contributes confounder information without adding outcome information beyond T_C, representations with more non-causal leakage incur a larger penalty. The purely causal representation $g(T) = T_C$ achieves the highest score among the tested representations.

After the non-causal component has been removed, the remaining question is whether the optimal representation should preserve all of T_C or compress it. To model lossy compression analytically, consider the stochastic representation channel

$$G = T_C + \epsilon_G, \qquad \epsilon_G \sim \mathcal{N}(0, \sigma_G^2),$$

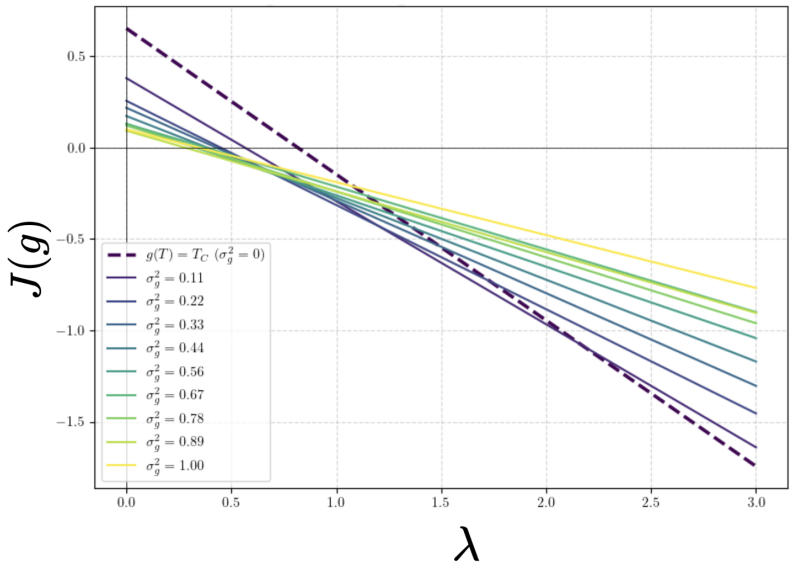

Figure 5: **Lossless regime and critical penalty weight.** The objective $J(g)$ is plotted as a function of λ for stochastic compressed representations $G = T_C + \epsilon_G$, where $\epsilon_G \sim \mathcal{N}(0, \sigma_G^2)$. The dashed curve corresponds to the uncompressed causal representation $\sigma_G^2 = 0$. For small λ, preserving the full causal component is optimal; for larger λ, compression can become preferable because the reduction in confounder information outweighs the utility loss.

with ϵ_G independent of all other variables. This can be viewed as a Gaussian Markov kernel from T_C to G, where σ_G^2 controls the amount of compression. The penalty is

$$P(\sigma_G^2) := I(G; Z) = \frac{1}{2} \log \left(1 + \frac{\alpha^2 \sigma_Z^2}{\sigma_C^2 + \sigma_G^2} \right),$$

and the utility is

$$U(\sigma_G^2) := I(G; Y \mid Z) = \frac{1}{2} \log \left(\frac{\rho^2 \sigma_C^2 + \sigma_Y^2}{(\rho^2 \sigma_C^2 + \sigma_Y^2) - \frac{\rho^2 \sigma_C^4}{\sigma_C^2 + \sigma_G^2}} \right).$$

Both terms decrease as σ_G^2 increases: compression removes information about the confounder, but it also removes predictive information about the outcome. The objective is

$$J(\sigma_G^2) = U(\sigma_G^2) - \lambda P(\sigma_G^2).$$

At the uncompressed boundary $\sigma_G^2 = 0$,

$$\left. \frac{dP}{d\sigma_G^2} \right|_{\sigma_G^2 = 0} = -\frac{\alpha^2 \sigma_Z^2}{2\sigma_C^2 (\sigma_C^2 + \alpha^2 \sigma_Z^2)},$$

and

$$\left. \frac{dU}{d\sigma_G^2} \right|_{\sigma_G^2 = 0} = -\frac{\rho^2}{2\sigma_Y^2}.$$

Thus the uncompressed representation is locally optimal whenever

$$\left. \frac{dJ}{d\sigma_G^2} \right|_{\sigma_G^2 = 0} = \left. \frac{dU}{d\sigma_G^2} \right|_{\sigma_G^2 = 0} - \lambda \left. \frac{dP}{d\sigma_G^2} \right|_{\sigma_G^2 = 0} \leq 0.$$

Equivalently,

$$\lambda \leq \lambda_{\text{crit}} := \frac{\left| \frac{dU}{d\sigma_G^2} \right|_{\sigma_G^2 = 0}}{\left| \frac{dP}{d\sigma_G^2} \right|_{\sigma_G^2 = 0}}.$$

This threshold is the ratio between the marginal utility loss and the marginal penalty reduction at zero compression. Figure 5 visualises this trade-off: for small λ, preserving the full causal component is optimal; for larger λ, the penalty becomes sufficiently expensive that a noisy compressed representation can achieve a higher objective value. This is the Gaussian analogue of the lossless regime in Proposition 3.

D NCE-CLUB Upper Bound and Critic Interpretation

To construct a tractable surrogate for the disentanglement objective

$$J(g) = I(g(T); Y \mid Z) - \lambda I(g(T); Z),$$

we require an upper bound on the penalty term $I(g(T); Z)$.

General NCE-CLUB bound. We use the NCE-CLUB estimator Liang et al. [2023]. For a pair of random variables (X_1, X_2), NCE-CLUB provides the upper bound

$$I(X_1; X_2) \leq I_{\text{NCE-CLUB}}(X_1; X_2),$$

where, for the optimal critic $f^\star$,

$$I_{\text{NCE-CLUB}}(X_1; X_2) := \mathbb{E}_{(x_1, x_2^+) \sim p(x_1, x_2)}\big[f^\star(x_1, x_2^+)\big] - \mathbb{E}_{x_1 \sim p(x_1),\, x_2^- \sim p(x_2)}\big[f^\star(x_1, x_2^-)\big].$$

The first expectation is over joint, or positive, pairs; the second is over independently sampled, or negative, pairs.

Application to the penalty term. Applying this construction to the penalty with

$$X_1 = g(T), \qquad X_2 = Z,$$

gives

$$I(g(T); Z) \leq I_{\text{NCE-CLUB}}(g(T); Z).$$

In practice, the optimal critic $f^\star$ is unknown, so we use a learned critic d_ϕ and obtain the plug-in estimator

$$\begin{aligned} I_{\text{NCE-CLUB}}(g(T); Z; d_\phi) := \mathbb{E}_{p(g(T), z)}\big[d_\phi(g(T), z)\big] \\ - \mathbb{E}_{p(g(T))p(z)}\big[d_\phi(g(T), z)\big]. \end{aligned}$$

This is the quantity used as the penalty estimator in the main text.

Interpretation of the critic d_ϕ. The optimal critic corresponds to the log-density ratio

$$d^\star(g, z) = \log \frac{p(z \mid g)}{p(z)} = \log p(z \mid g) - \log p(z),$$

which is intractable because the true conditional $p(z \mid g)$ is unknown. We therefore approximate it with a learnable critic $d_\phi(g, z)$, which can be interpreted as a variational estimate

$$d_\phi(g_\theta(T), z) \approx \log q_\phi(z \mid g_\theta(T)) - \log p(z),$$

where $q_\phi(z \mid g)$ is a learned conditional model.

The critic is trained to distinguish positive pairs

$$(g, z) \sim p(g, z)$$

from negative pairs

$$(g, z) \sim p(g)p(z).$$

This training aims to move d_ϕ toward the optimal log-ratio and thereby tighten the plug-in estimate.

Adversarial training dynamics. The optimisation alternates between critic fitting and encoder updates. The critic parameters ϕ are trained to maximise the InfoNCE objective, approximating the optimal critic for the current representation. Then, holding the critic fixed, the encoder parameters θ are trained to minimise the NCE-CLUB penalty together with the utility loss. This encourages representations that remain predictive of Y while discouraging dependence on Z. The gradient-reversed InfoNCE implementation used in our experiments is described in Section 2.4.

E Synthetic SCM: experimental details

We generate 1,000 training samples, and 1,000 validation samples from the following linear additive SCM with $d = 10$, $d_c = 5$:

$$\begin{aligned} Z &\sim \mathcal{N}(0, I_d), \\ T_C &= Z_{0:d_c} + \epsilon_{T_C}, & \epsilon_{T_C} &\sim \mathcal{N}(0, I_{d_c}), \\ T_{nC} &= Z_{d_c:d} + \epsilon_{T_{nC}}, & \epsilon_{T_{nC}} &\sim \mathcal{N}(0, I_{d-d_c}), \\ T &= [T_C, T_{nC}], \\ Y &= \mathbf{1}^\top T_C + \mathbf{1}^\top Z + \epsilon_Y, & \epsilon_{Y,i} &\overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma_Y^2). \end{aligned}$$

The predictor is a linear model $\hat{Y} = w_Z^\top Z + w_T^\top T + b$. We take the treatment representation to be the per-coordinate contribution of T to the prediction, $g_T = T \odot w_T \in \mathbb{R}^{10}$, where w_T is the treatment slice of the predictor's own weight vector. The MI critic therefore decorrelates Z from the components of T that the predictor actually uses. Under gradient reversal this pushes w_T toward the causal coordinates of T. The critic itself is bilinear: two bias-free linear projections $W_g, W_z \in \mathbb{R}^{10\times 10}$ to a shared embedding space, with InfoNCE scores given by the full $B \times B$ matrix $S = (W_g g_T)(W_z Z)^\top$ over a minibatch of size B. We train with lr 3×10^{-4}, batch size 64, 4,000 epochs, and GRL schedule with $\lambda = 0$ for epochs 0–200, cubic ramp $0 \to \lambda_{\max} = 0.5$ over epochs 200–1000, held constant thereafter.

F Semi-Synthetic KuaiRand: experimental details

We use the KuaiRand-Pure Gao et al. [2022] (Zenodo 10439422) dataset with 27,285 users, 7,583 items, and ∼1.186M distinct user-video pairs from the randomized-exposure log. KuaiRand provides 30 user features, of which we keep 24 including four counts, two binary indicators, and 18 anonymized category codes. We drop the four bucketed versions of the counts, the categorical activity label, and one binary indicator that is constant in KuaiRand-Pure. For simplicity when generating synthetic outcomes we treat each anonymized category code as numerical. KuaiRand also provides 62 video features, of which we keep 54 including duration, width, height, and 51 video statistics. We drop identifiers, categorical features, upload date, and constant features. We log transform counts, duration, video statistics, and then standardize them. We standardize the numerical user features.

Let Z denote the 24 user features and T_C the 54 video features. We generate a synthetic click outcome from a user term $h_u(Z)$, video term $h_v(T_C)$, and user-video interaction term $h_{uv}(Z, T_C)$ as

$$Y \sim \text{Bernoulli}\left(\text{sigmoid}\left(\text{std}\left(4\, h_u(Z) + h_v(T_C) + h_{uv}(Z, T_C)\right) + b\right)\right)$$

where std corresponds to centering and scaling. The intercept b calibrates the mean click probability to the actual click rate in the randomized log. We define the user, video, and user-video interaction terms as follows. Let $q_1, q_2 \in \mathbb{R}^{24}$, $q_3 \in \mathbb{R}^{54}$, $Q_1 \in \mathbb{R}^{24\times 16}$, and $Q_2 \in \mathbb{R}^{54\times 16}$ be fixed random projections. We define

$$\begin{aligned} h_u(Z) &= \text{std}\left(\tanh\big(\text{std}(Z^\top q_1)\big)\tanh\big(\text{std}(Z^\top q_2)\big)\right), \\ h_v(T_C) &= \text{std}(T_C^\top q_3), \\ h_{uv}(Z, T_C) &= \text{std}\left(\frac{1}{16}\sum_{j=1}^{16}\tanh\big((Z^\top Q_1)_j\big)\tanh\big((T_C^\top Q_2)_j\big)\right). \end{aligned}$$

The entries of each projection are independent zero-mean Gaussian samples with variance equal to the inverse of its input dimension. To be clear, the data generation process above is not intended to be a realistic model of user behaviour. It provides a nonlinear outcome mechanism with known causal structure built from real KuaiRand features.

To test our method, we introduce a non-causal treatment component $T_{nC} \in \{1, \ldots, 16\}$, which does not impact the outcome, but is related to Z and can be exploited by the model when making predictions. We append this non-causal shortcut feature to the observed treatment input $T = (T_C, T_{nC})$. In the

source distribution, T_{nC} is strongly related to the user response term. Let $Q(Z)$ be the quantile bin of $h_u(Z)$, then

$$p(T_{nC} = t \mid Z) = \frac{\exp\left(\alpha \mathbf{1}\left[t = Q(Z)\right]\right)}{\exp(\alpha) + 15}.$$

The parameter α controls the shortcut strength, and we choose α so that the mutual information is $I(T_{nC}; Q \mid V) = 2$ nats, where V is the video ID. For the target distribution, we instead set $\alpha = 0$, sampling the non-causal feature uniformly and removing the Z shortcut. Source and target test sets contain the same users, videos, features, and synthetic clicks, only T_{nC} changes. We encode the code as 16 one-hot values scaled by 3, so the treatment encoder receives $54 + 16 = 70$ inputs.

We train on 70% of the data, use 15% as our validation set for early stopping, and the remaining 15% as the test set for calculating source and target AUC and ranking metrics. Our model is a DeepFM Guo et al. [2017] backbone with 16-dimensional learned user-ID and video-ID embeddings. The treatment encoder maps $70 \to 32 \to 16$ with a ReLU hidden layer. Appending the video-ID embedding gives a 32-dimensional treatment representation. Appending the user-ID embedding to the 24 user features gives a 40-dimensional user representation. The DeepFM network has hidden sizes $[32, 16]$, dropout 0.2, and FM embedding dimension 8. The CRL variant uses the same predictor and adds an InfoNCE critic. The predictor architecture and capacity are identical across configurations. Each critic input is mapped through a two-layer projection with widths 64 and 32, followed by a dot-product score. Gradient reversal acts on the treatment representation. The user representation is detached from the critic loss. The reversal weight λ is zero for 25 steps, rises linearly to 0.3 over the next 50 steps, and then remains fixed. We train for four epochs with Adam, learning rate 2×10^{-3}, weight decay 10^{-6}, batch size $4{,}096$, and gradient-norm clipping at 1. We evaluate source-validation AUC every 50 steps and retain the best checkpoint.

When applying DCCL Zhao et al. [2023] to our KuaiRand setup, we split each 16-dimensional user and video ID embedding equally into interest and conformity components, preserving the predictor's total capacity. Two popularity-weighted contrastive losses supplement the prediction loss, using clicked examples as anchors and up to 128 in-batch negatives. Both auxiliary losses use weight 0.01.

As well as source and target AUC, in Table 1 we report target Recall@5 and target NDCG@5 [see eg. Zhang et al., 2023, Deng et al., 2024, who report similar metrics on KuaiRand]. For each user in the test split we rank the items they were exposed to by predicted click probability. Recall@K is the fraction of that user's positives that appear in the top-K, and NDCG@K uses binary relevance with discount $1/\log_2(i+1)$ at rank i and IDCG computed over the user's true positives. Both are averaged over users with at least one positive and at least five exposed items.